

Attachment 3, RPMS-ENGG-PRA-GE-041, Rev 01

Reliance

Chemicals and Materials Limited

BECHTEL

PMC: BECHTEL ENERGY

Vendor Document Review Sheet (DECAL)

GMM

Pfaudler

thyssenkrupp

PROJECT TITLE:

VINYLS PROGRAM - 1000 KTA VINYL CHLORIDE MONOMER (VCM)

CLIENT:

Reliance Chemicals and Materials Limited

LOCATION:

Nagothane Manufacturing Division, India

CONTRACTOR:

thyssenkrupp Uhde India Private Limited

tkUIPL Contract No.:

66-6785

VENDOR NAME:

GMM PFAUDLER LTD

DOCUMENT TITLE:

Inspection & Test Plan

VENDOR DOCUMENT NO:

ITP-HEV000159-COMMON

VENDOR REVISION NO:

00

thyssenkrupp

Uhde

Acceptance Code

2

APPROVED AS NOTED /
RELEASE FOR CONSTRUCTION

This approval does not release the vendor / supplier from full responsibility for his design and fabrication.

Date: March 22, 2025

Name: Patel, Naitikkumar

ANA

VENDOR DOCUMENT REVIEW FEEDBACK

Permission to proceed does not constitute acceptance or approval of Design Detail, Calculations, Analysis, Test Methods or Materials developed OR selected by Vendor, and does NOT relieve Vendor from FULL compliance with Contractual Obligations

☐

1

Approved

☐

2

Approved with comments. Work may proceed.

☐

3

Revise and Resubmit. Work shall not proceed

☐

4

Retained for Information

☐

5

Cancelled / Superseded

☐

F

Final Certified

By:

Date:

Discipline:

Mechanical

MOC Number:

N/A

Jl Number

N23P02

Equipment No.

Add design data here like what is followed for previous Order of VCM & EDC

MRQ Number

Equipment Name:

Program No

Tag No.:

NP3701

Bottom Dish thickness

MDMT

Top Dish thickness

Main shell thickness

Impact

PWHT

Unit No.(Plant WBS)

Cat. No

Seq. No.

Rev No

-

0

2

1

-

A

0

5

-

0

0

1

-

0

0

Page 1 of 1

Inspection and Test Plan for QUENCH SEPERATOR



CUSTOMER :- Reliance Chemicals & Materials Limited	ITP NO:	HEV000159-01&04-10, Rev.0, Date: 20.02.2025
PROJECT:- 1000 KTA VINYL CHLORIDE MONOMER (VCM)	GMM SO No:	HEV000159-01 to 04
Detail Engineering Contractor(DEC):- thyssenkrupp Uhde India Private Limited (tkUIPL)	Equipment Name:	QUENCH SEPERATOR
Applicable Code:- ASME VIII DIV.1. 2023 ED	Equipment TAG No.	MV-PV021-V141004A/B, MV-PV022-V143004A/B
Drq No. HEV000159-01 to 04-01	Add PR Doc No:	PO No: 36C/240840258
Mechanical Datasheet No. NP3701-MVVD-PV021-141004	SQS Level / Criticality Rating : 1	
Purchase Requisition: MT-N23P02-D11NPV-MVV002-TIN-P-01		

SR. NO.	ITEM DESCRIPTION	CHARACTERISTIC CHECK	EXTENT OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS/ STANDARDS	FORMAT OF RECORDS	AGENCY				REMARKS
							GMM	TPIA	tkUIPL	PMC/ RCML	
1.0	Document Approvals										
1.1	Design calculation and Drawing	Review	100%	Applicable Code & Standards, Data sheet, MR etc	Applicable Code & Standards, Data sheet, MR etc	Approved Drawing & design calculations	P	-	A		
1.2	Inspection and Test Plan	Review	100%	Applicable Code & Standards, Data sheet, MR, PO	Applicable Code & Standards, Data sheet, MR, PO	Approved ITP	P	R	A		
1.3	Welding procedure specifications and Procedure Qualification Record including weld repair procedure, Weld Map etc	Welding parameters, Chemical analysis, Mechanical properties, Heat Treatment cycles, NDE results etc (as applicable)	100%	Specification, Drawing, Applicable Code & Standards RPMS-ENGG-SPC-MT-004	Specification, Drawing, Applicable Code & Standards RPMS-ENGG-SPC-MT-004	WPS, PQR, WPQ, Weld Map.	P	R/W*	R		* R - Review for existing Qualification, W - Witness for New Qualification under RCML Approved TPIA, Welder Performance Qualification (WPQ) records shall be reviewed by TPIA Existing WPS /PQR shall be under RIL Approved TPIA or else new qualification is required.
1.4	NDE Procedure - RT, UT, MT, PT Load test procedure for Manhole davit &	Procedure	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Approved procedure	P	R	R		NDE Procedure should be approved by ASNT Level III.
1.5	NDE Operator & Personnel Qualification	Qualification, Validity etc	100%	Specification, Applicable Code & Standards, Standards, Society	Specification, Applicable Code & Standards, Standards, Society	Certificate	P	R	R		NDE Operators shall be ASNT Level II minimum
1.6	Post forming Heat treatment Procedure	Procedure	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Approved procedure	P	R	R		
1.7	Leak Test (Hydro & Pneumatic) Procedure	Procedure	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Approved procedure	P	R	A		
1.8	PMI, Hardness Test Procedure etc	Procedure	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Approved procedure	P	R	A		
1.9	Surface Preparation & Painting	Procedure	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Approved procedure	P	R	A		

Inspection and Test Plan for QUENCH SEPERATOR



CUSTOMER :- Reliance Chemicals & Materials Limited					ITP NO:		HEV000159-01&04-10, Rev.0, Date: 20.02.2025				
PROJECT:- 1000 KTA VINYL CHLORIDE MONOMER (VCM)					GMM SO No:		HEV000159-01 to 04				
Detail Engineering Contractor(DEC):- thyssenkrupp Uhde India Private Limited (tkUIPL)					Equipment Name:		QUENCH SEPERATOR				
Applicable Code:- ASME VIII DIV.1. 2023 ED					Equipment TAG No.		MV-PV021-V141004A/B, MV-PV022-V143004A/B				
Drq No. HEV000159-01 to 04-01					PO No:		36C/240840258		SQS Level / Criticality Rating : 1		
Mechanical Datasheet No. NP3701-MVVD-PV021-141004					Purchase Requisition: MT-N23P02-D11NPV-MVV002-TIN-P-01						
SR. NO.	ITEM DESCRIPTION	CHARACTERISTIC CHECK	EXTENT OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS/ STANDARDS	FORMAT OF RECORDS	AGENCY				REMARKS
							GMM	TPIA	tkUIPL	PMC/ RCML	
1.10	Draining & Drying, Desiccant Protection Procedure etc	Procedure	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Approved procedure	P	R	A		
1.11	Packing, Shipping, Handling & Storage Procedure	Procedure	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Approved procedure	P	R	A		
1.12	Pre Inspection meeting	Discussion of ITP, Specification requirements, responsibilities, communication protocol etc	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	PIM MOM	H	H	H		PIM shall be conducted after first review of ITP, Drawings & Weld book
2.0	Material Identification										
2.1	Identification and stamp transfer of Pressure parts	Visual, Dimension, Mechanical, Chemical, Heat treatment condition, NDT, Approved Manufacturer etc (as applicable).	100%	Specification, Drawing, Applicable Code & Standards, Approved TDC	Specification, Drawing, Applicable Code & Standards, Approved TDC	Identification Report and MTC's (EN 10204-3.1/3.2)	P	W	R	R	Add stage PMI of SS & LAS (as applicable)-Alloying Elements - Cr, Moly, Ni (As applicable)- Applicable for fasteners and gaket-W by TPIA
2.2	Identification and stamp transfer of parts directly welded to pressure part	Visual, Dimension, Mechanical, Chemical, Heat treatment condition as per Material specification	100%	Material Specification, Drawing, Applicable Code & Standards, Approved TDC	Material Specification, Drawing, Applicable Code & Standards, Approved TDC	Identification Report and MTC's	P	W	R	R	
2.3	Identification and stamp transfer of Non pressure parts and Load bearing parts.	Visual, Dimension, Mechanical, Chemical, Heat treatment condition as per Material specification	100%	Material Specification, Drawing, Applicable Code & Standards	Material Specification, Drawing, Applicable Code & Standards	Identification Report and MTC's	P	R	R	R	
2.4	Identification and stamp transfer of Wetted parts	Visual, Dimension, Mechanical, Chemical, Heat treatment condition, Corrosion test as per Material specification	100%	Material Specification, Drawing, Applicable Code & Standards	Material Specification, Drawing, Applicable Code & Standards	Identification Report and MTC's	P	W	R	R	Wetted parts shall have same properties as per Pressure
2.6	Weld Consumables	Chemical, Mechanical properties, Impact, Approved Manufacturer etc (as applicable)	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Batch TC	P	R	R	R	Reputed make duly approved by TPIA like TUV, SGS, LR &BV. Low Hydrogen electodes & flux having maximum 10mi/100gm shall be used. R - Review of TC.
3.1	Formed Component - Dished End Inspection										

Inspection of Critical bought out items to be inspected by TPIA at Sub vendor's work
(Sourced within India)

Visual, Dimension, NDT, Chemical, Mechanical properties, Impact, Approved Manufacturer etc (as applicable)

W By TPIA

TYP comment- TPIA stages shall be strictly as per IITP. Most of stages are marked as R by GMM . same is not acceptable. Revise



Inspection and Test Plan for QUENCH SEPERATOR

CUSTOMER :- Reliance Chemicals & Materials Limited					ITP NO:		HEV000159-01&04-10, Rev.0, Date: 20.02.2025				
PROJECT:- 1000 KTA VINYL CHLORIDE MONOMER (VCM)					GMM SO No:		HEV000159-01 to 04				
Detail Engineering Contractor(DEC):- thyssenkrupp Uhde India Private Limited (tkUIPL)					Equipment Name:		QUENCH SEPERATOR				
Applicable Code:- ASME VIII DIV.1. 2023 ED					Equipment TAG No.		MV-PV021-V141004A/B, MV-PV022-V143004A/B				
Drq No. HEV000159-01 to 04-01					PO No:		36C/240840268		SQS Level / Criticality Rating : 1		
Mechanical Datasheet No. NP3701-MVVD-PV021-141004					Purchase Requisition: MT-N23P02-D11NPV-MVV002-TIN-P-01						
SR. NO.	ITEM DESCRIPTION	CHARACTERISTIC CHECK	EXTENT OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS/ STANDARDS	FORMAT OF RECORDS	AGENCY				REMARKS
							GMM	TPIA	tkUIPL	PMC/ RCML	
3.1	Plate to plate setup and welding of Long seam of RHS and LHS dish end	Visual, Dimension, Ovality, Mismatch, Weld Edge preparation, Geometry, Alignment, PT of WEP	100%	Drawing, Specification, Applicable Code & Standards	Drawing, Specification, Applicable Code & Standards	Inspection Report	P	RW	R	R	
3.2	Back chipping and PT on Gauged area of Long seam of RHS and LHS dish end	PT of Back Chip, PT of Inside and outside Kunckle Portion	100%	Drawing / NDE procedure	Drawing / NDE procedure	NDE Report	P	R	R	R	
3.3	Heat treatment (Normalising) of RHS and LHS dish end along with Material test coupon	Heat treatment Cycle.	100%	Drawing, Specification, Applicable Codes & Standards, HT Procedure	Drawing, Specification, Applicable Codes & Standards, HT Procedure	Heat treatment Chart	P	R	R	R	
3.4	Inspection of Dish Head after forming & Heat treatment Template check on layout	Dimensions (Circumference, Ovality, Height, Knuckle radius, crowning, straight face, Thickness check by Ultrasonic Thickness gauge), Profile check by Template	100%	Drawing, Specification, Applicable Codes & Standards	Drawing, Specification, Applicable Codes & Standards	Inspection Report	P	W	R	R	
3.5	NDE Examination after forming & Heat treatment Tests in Dish Heads shall be Radiographed after forming & subsequent Heat treatment (if applicable)	PT of knuckle surface, SF from inside & outside	100%	Drawing, Specification, Applicable Codes & Standards, NDE Procedure	Drawing, Specification, Applicable Codes & Standards, NDE Procedure	NDE reports, RT Films as applicable	P	W	R	R	
		RT of Long Seam					P	R	R	R	RT films review
3.6	Simulation test coupon Testing of Heat treated components (Dished end)	Mechanical Test - Tensile	100%	Drawing, Specification, Applicable Procedure, codes & standards	Drawing, Specification, Applicable Procedure, codes & standards	Approved Inhouse or NABL approved Laboratory Test reports	P	W	R	R	
4.0	In-process										
4.1	Weld Edge Preparation and Setup of Long seam of Main Shell and Jacket	Visual, Dimension, Ovality, Mismatch, Weld Edge preparation, Geometry, Alignment, PT of WEP, Profile check by Template (If applicable)	100%	Drawing, Specification, Applicable Code & Standards	Drawing, Specification, Applicable Code & Standards	Inspection Report	P	RW	R	R	W for first stage and RW for balance joints, Each vessel RW for any one Tag Jacket shell along with PTC as per IBR regulation

1) Weld seams in Dish Heads shall be Radiographed & DP/MT tested after forming & subsequent Heat treatment (as applicable)

W for first stage and RW for balance joints, Each vessel

Inspection and Test Plan for QUENCH SEPERATOR



CUSTOMER :- Reliance Chemicals & Materials Limited					ITP NO:		HEV000159-01&04-10, Rev.0, Date: 20.02.2025				
PROJECT:- 1000 KTA VINYL CHLORIDE MONOMER (VCM)					GMM SO No:		HEV000159-01 to 04				
Detail Engineering Contractor(DEC):- thyssenkrupp Uhde India Private Limited (tkUIPL)					Equipment Name:		QUENCH SEPERATOR				
Applicable Code:- ASME VIII DIV.1. 2023 ED					Equipment TAG No.		MV-PV021-V141004A/B, MV-PV022-V143004A/B				
Drq No. HEV000159-01 to 04-01					PO No:		36C/240840258			SQS Level / Criticality Rating : 1	
Mechanical Datasheet No. NP3701-MVVD-PV021-141004					Purchase Requisition: MT-N23P02-D11NPV-MVV002-TIN-P-01						
SR. NO.	ITEM DESCRIPTION	CHARACTERISTIC CHECK	EXTENT OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS/ STANDARDS	FORMAT OF RECORDS	AGENCY				REMARKS
							GMM	TPIA	tkUIPL	PMC/ RCML	
4.2	Back chipping and PT on Gauged area and final welding of Long seam	PT of Back Chip (where applicable)	100%	Drawing / NDE procedure	Drawing / NDE procedure	NDE Report	P	RW	R	R	
4.3	RT of butt welds of long seam welds	RT	100%	Drawing, Specification, Applicable Codes & Standards, NDE Procedure	Drawing, Specification, Applicable Codes & Standards, NDE Procedure	NDE Reports, RT Films as applicable	P	R	R	R	RT - Film review,
4.4	Weld Edge Preparation and Setup of Cir-seam of main shell, jacket & Pipe to flange	Visual, Dimension, Ovality, Mismatch, Weld Edge preparation, Geometry, Alignment, PT of WEP, Profile check by Template	100%	Drawing, Specification, Applicable Code & Standards	Drawing, Specification, Applicable Code & Standards	Inspection Report	P	RW	R	R	W for first stage and RW for balance RW for any one Tag
4.5	Back chipping and PT on Gauged area and final welding of Cir-seam of main shell, jacket & Pipe to flange	PT of Back Chip (where applicable)	100%	Drawing / NDE procedure	Drawing / NDE procedure	NDE Report	P	RW	R	R	
4.6	RT of butt welds of Cir-seam welds	RT	100%	Drawing, Specification, Applicable Codes & Standards, NDE Procedure	Drawing, Specification, Applicable Codes & Standards, NDE Procedure	NDE Reports, RT Films as applicable	P	R	R	R	RT - Film review,
4.7	Weld Edge Preparation and Setup and final welding of Nozzle to shell / dish	Visual, Dimension, Mismatch, Weld Edge preparation, Geometry, Alignment, PT of WEP	100%	Drawing, Specification, Applicable Code & Standards	Drawing, Specification, Applicable Code & Standards	Inspection Report	P	RW	R	R	RW for any one Tag
4.8	Back chipping and PT on Gauged area and final welding of Nozzle to shell / dish	PT of Back Chip (where applicable)	100%	Drawing / NDE procedure	Drawing / NDE procedure	NDE Report	P	R	R	R	
4.9	Visual inspection of Main shell from outside before fit up of Jacket.	visual inspection	100%	Drawing	Drawing	Inspection Report	P	W	R	R	
4.10	Fit up Jakcet on main shell along with closure ring	Visual, Dimension, Ovality, Mismatch, Weld Edge preparation, Geometry, Alignment, NDE etc	100%	Drawing, Specification, Applicable Code & Standards	Drawing, Specification, Applicable Code & Standards	Inspection Report	P	W	R	R	

Trial Assembly & Fitment check of Internals like Distributors, Dip pipes, Demisters/Coalescer Pads, etc (internals which are in the supply scope of GMM)-W by TPIA

Internals shall be Inspected before Fitment /Trial assembly.
Visual inspection will be offered at Final dimension inspection.

UT at area of Shell/Head/Skirt etc where lifting lugs, tailing lugs, trunnions etc as applicable to welded-W by TPIA

UT of Reinforcement pads beneath lifting lugs/trunnions to verify absence of lamination in the steel-W by TPIA

UT for trunnion to shell joints for vessels with lifting weight greater than 100 MT-W by TPIA

QUENCH SEPERATOR



CUSTOMER :- Reliance Chemicals & Materials Limited					ITP NO:		HEV000159-01&04-10, Rev.0, Date: 20.02.2025				
PROJECT:- 1000 KTA VINYL CHLORIDE MONOMER (VCM)					GMM SO No:		HEV000159-01 to 04				
Detail Engineering Contractor(DEC):- thyssenkrupp Uhde India Private Limited (tkUIPL)					Equipment Name:		QUENCH SEPERATOR				
Applicable Code:- ASME VIII DIV.1. 2023 ED					Equipment TAG No.		MV-PV021-V141004A/B, MV-PV022-V143004A/B				
Drq No. HEV000159-01 to 04-01					PO No:		36C/240840258			SQS Level / Criticality Rating : 1	
Mechanical Datasheet No. NP3701-MVVD-PV021-141004					Purchase Requisition: MT-N23P02-D11NPV-MVV002-TIN-P-01						
SR. NO.	ITEM DESCRIPTION	CHARACTERISTIC CHECK	EXTENT OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS/ STANDARDS	FORMAT OF RECORDS	AGENCY				REMARKS
							GMM	TPIA	tkUIPL	PMC/ RCML	
4.11	Set up and welding of internal/external attachment	Dimension & visual	100%	Drawing	Drawing	Inspection Report	P	R	R	R	
4.12	Pneumatic test for RF Pad	Leak tightness	100%	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Inspection Reports	P	W	R	R	"e" value to be checked of Shell, Heads etc with Internal/ external Template (Applicable for equipment with External pressure/Full or Partial Vacuum) 2) Other Tolerances shall be strictly as per Appendix C of RPMS-ENGG-SPC-MS-002.
5.0	Final Inspection & NDE before PWHT										
5.1	Visual and Dimension inspection of Equipment before PWHT	Visual, Overall Dimensions, Elevation, Projection of Nozzles and attachments, Perpendicularity, Weld fillet size check, Alignment, compliance of all welding including internal, flatness check of Tray support ring, Bolting bar, Element support rings, Seal pan etc, Completeness check, Matching of , Functional test of Davit Arms, Hinge assembly etc	100%	Drawing, Specification, Applicable Code & Standards RPMS-ENGG-SPC-MS-002' Appendix C	Drawing, Specification, Applicable Code & Standards RPMS-ENGG-SPC-MS-002' Appendix C	Inspection & NDE Report	P	H	R	R	PI add separate Stages for SS Nozzles N03, N05, N06...etc welding/Buttering before PWHT
5.2	PT on all pressure bearing butt & fillet welds, load bearing welds (other attachment welds directly attached to pressure parts & non pressure lifting lug welds)	PT to check surface defect	100%	Drawing, Specification, NDE Procedure, Applicable Code & Standards	Drawing, Specification, NDE Procedure, Applicable Code & Standards	Inspection & NDE Report	P	W	R	R	PT, PMI Thk of buttering. Discuss with welding engineer & Design team. This was discussed long back to avoid PWHT of SS parts
5.3	NDE clearance of equipment prior to PWHT	NDE (PT/MT, UT & RT)	100%	Drawing, Specification, NDE Procedure, Applicable Code & Standards	Drawing, Specification, NDE Procedure, Applicable Code & Standards	Inspection Report & NDE Reports	P	H	R	R	Review of NDE reports & RT Summary,
5.4	PWHT of complete vessel	time-temp chart	100%	Drawing, Specification, PWHT Procedure, Applicable Code & Standards	Drawing, Specification, PWHT Procedure, Applicable Code & Standards	Inspection Report & NDE Reports	P	R	R	R	add stage for Main davit & man hole davit load test -W by TPIA Apply load as per drawing 100%
6.0	Inspection after PWHT										
6.1	Verification of material heat chart, NDE summary and stage inspection reports and released for hydro test	Verification of inspection report	100%	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Inspection Report	P	H	R	R	NDE before & after load test PT test of load bearing weld join-W by TPIA

"e" value to be checked of Shell, Heads etc with Internal/ external Template (Applicable for equipment with External pressure/Full or Partial Vacuum)
2) Other Tolerances shall be strictly as per Appendix C of RPMS-ENGG-SPC-MS-002.

PI add separate Stages for SS Nozzles N03, N05, N06...etc welding/Buttering before PWHT

PT, PMI Thk of buttering. Discuss with welding engineer & Design team. This was discussed long back to avoid PWHT of SS parts

add stage for Main davit & man hole davit load test -W by TPIA

Apply load as per drawing 100%

NDE before & after load test PT test of load bearing weld join-W by TPIA

Verification of material heat chart, NDE summary and stage inspection reports and released for PWHT

PMI of Material of SS component and fasteners, welds-W by TPIA

MT/PT on all pressure bearing butt (C-seam of fabricated pipe/pipe/pipe fittings) & fillet welds (Nozzle to main shell/dish end welds), load bearing welds (other attachment welds directly attached to pressure parts & non pressure lifting lug welds), Vessel to skirt support joint final run, temporary attachments & arc strikes after removal (as applicable)-W by TPIA

Platform and ladder Items Inspection stages ...???

inal visual inspection of platform and ladder in black condition

Visual and marking 100% Trial Assembly of Platform & Ladder (one of each type)

Final paint
DFT measurement, Visual, Pull of adhesion, adhesion

Inspection and Test Plan for QUENCH SEPERATOR



CUSTOMER :- Reliance Chemicals & Materials Limited					ITP NO:	HEV000159-01&04-10, Rev.0, Date: 20.02.2025					
PROJECT:- 1000 KTA VINYL CHLORIDE MONOMER (VCM)					GMM SO No:	HEV000159-01 to 04					
Detail Engineering Contractor(DEC):- thyssenkrupp Uhde India Private Limited (tkUIPL)					Equipment Name:	QUENCH SEPERATOR					
Applicable Code:- ASME VIII DIV.1. 2023 ED					Equipment TAG No.	MV-PV021-V141004A/B, MV-PV022-V143004A/B					
Drq No. HEV000159-01 to 04-01					PO No:	36C/240840258		SQS Level / Criticality Rating : 1			
Mechanical Datasheet No. NP3701-MVVD-PV021-141004					Purchase Requisition: MT-N23P02-D11NPV-MVV002-TIN-P-01						
SR. NO.	ITEM DESCRIPTION	CHARACTERISTIC CHECK	EXTENT OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS/ STANDARDS	FORMAT OF RECORDS	AGENCY				REMARKS
							GMM	TPIA	tkUIPL	PMC/ RCML	
6.2	RT of pressure butt weld after PWHT	RT to check weld soundness	100%	Drawing, Specification, Applicable Codes & Standards, NDE Procedure	Drawing, Specification, Applicable Codes & Standards, NDE Procedure	NDE Reports, RT Films as applicable	P	R	R	R	RT - Film review,
6.3	PT of temporary attachment removal area, load bearing welds.	PT to check surface defect	100%	Drawing, Specification, NDE Procedure, Applicable Code & Standards	Drawing, Specification, NDE Procedure, Applicable Code & Standards	Inspection & NDE Report	P	W			Weld deposits shall be hardness tested, where required, on the side contacted by the process, whenever possible. For vessel or tank butt welds where hardness testing is required, a minimum of one location per weld seam shall be hardness tested. Unless otherwise specified by the user, one hardness test should be
6.4	Hardness check on pressure welds and HAZ	Max hardness 200BHN, one spot per each weld joint	100%	Drawing, Specification, Applicable Codes & Standards, Hardness Procedure	Drawing, Specification, Applicable Codes & Standards, Hardness Procedure	Inspection Report	P	W	R	R	made for each 3 m (10 ft) of weld seam. In addition, one hardness test shall be made on each nozzle flange-to-neck and nozzle neck-to-shell/head weld. Each unique welding procedure used shall be hardness tested. 2.27.5 When hardness testing of welds is required, fillet weld deposit hardness testing should be done when access is feasible.
6.5	Hydrostatic test for Equipment	Leak Tightness, Pressure Gauge (Min 2nos), Range, Leakage, Pressure drop, (Holding Time - 60minutes minimum)	100%	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Inspection Report	P	H	R	R	Potable quality with chloride content less than 250 ppm. Service gaskets shall be used during hydrotest for gasketed joints.
6.6	Draining & Drying	Cleanliness, etc	100%	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Inspection Report	P	W	R	R	
6.7	Surface preparation of External CS surface	Blasting media, Surface profile, Salt contamination test as per RIL specification etc	100%	Specification, Drawing, Applicable Code & Standards, Approved Procedure RPMS-ENGG-SPC-PP-014	Specification, Drawing, Applicable Code & Standards, Approved Procedure RPMS-ENGG-SPC-PP-014	Inspection Report	P	R	R	R	Pickling and passivation of SS parts & welds - - Visual W by TPIA

Inspection and Test Plan for QUENCH SEPERATOR



CUSTOMER :- Reliance Chemicals & Materials Limited					ITP NO:		HEV000159-01&04-10, Rev.0, Date: 20.02.2025				
PROJECT:- 1000 KTA VINYL CHLORIDE MONOMER (VCM)					GMM SO No:		HEV000159-01 to 04				
Detail Engineering Contractor(DEC):- thyssenkrupp Uhde India Private Limited (tkUIPL)					Equipment Name:		QUENCH SEPERATOR				
Applicable Code:- ASME VIII DIV.1. 2023 ED					Equipment TAG No.		MV-PV021-V141004A/B, MV-PV022-V143004A/B				
Drq No. HEV000159-01 to 04-01					PO No:		36C/240840258			SQS Level / Criticality Rating : 1	
Mechanical Datasheet No. NP3701-MVVD-PV021-141004					Purchase Requisition: MT-N23P02-D11NPV-MVV002-TIN-P-01						
SR. NO.	ITEM DESCRIPTION	CHARACTERISTIC CHECK	EXTENT OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS/ STANDARDS	FORMAT OF RECORDS	AGENCY				REMARKS
							GMM	TPIA	tkUIPL	PMC/ RCML	
6.8	Painting of External CS surface	DFT, Pull of adhesion test, Shade verification, as per RIL specification etc	100%	Specification, Drawing, Applicable Code & Standards, Approved Procedure RPMS-ENGG-SPC-PP-014	Specification, Drawing, Applicable Code & Standards, Approved Procedure RPMS-ENGG-SPC-PP-014	Inspection Report & Paint TC	P	W	R	R	Witness by TPI - Final DFT Witnessed- 1) DFT, 2) Pull of Adhesion test 3) Adhesion, Visual
6.9	Final Inspection including Name Plate, Marking and Stamping check.	Verification, Completion, Code compliance, compliance with drawing.	100%	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Inspection Report	P	H	R	R	
6.10	Packing & Preservation	Desiccant Protection -Silica gel bags (0.5kg/m3), Caution, CG, Sling marking, Fastener rust preventive application, 100% Nozzle bolting as per flange standard, etc (as applicable)	100%	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Specification, Drawing, Applicable Code & Standards, Approved Procedure	Inspection Report	P	W	R	R	Jacket shall be purged with nitrogen 0.3 kg/cm2 minimum pressure Silica gel bags shall be placed inside Main shell
6.11	Inspection of Spares	Visual, Dimensional, PMI & Quantity verification	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Inspection Report	P	H	R	R	
6.12	Sign off ITP	Compliance to ITP, Drawings, Requisition & Specifications	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Inspection Report	P	H	R	R	
6.13	Stamping & Issue of Inspection Release Note	Compliance to ITP, Drawings, Requisition & Specifications	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	Inspection Release Note	P	H	H	R	
6.14	Engineering Clearance (EC) and Supplier Quality (SQ) clearance for shipment Release.	Compliance to ITP, Drawings, Requisition & Specifications	100%	Specification, Drawing, Applicable Code & Standards	Specification, Drawing, Applicable Code & Standards	EC & SQ Clearance	H	H	H	H	
6.15	Manufacturer record Book (MRB)	Review of MRB against approved MRB index	100%	Review of MRB	Review of MRB	MRB dossier	H	R	R	R	

Inspection and Test Plan for QUENCH SEPERATOR



CUSTOMER :- Reliance Chemicals & Materials Limited					ITP NO:		HEV000159-01&04-10, Rev.0, Date: 20.02.2025				
PROJECT:- 1000 KTA VINYL CHLORIDE MONOMER (VCM)					GMM SO No:		HEV000159-01 to 04				
Detail Engineering Contractor(DEC):- thyssenkrupp Uhde India Private Limited (tkUIPL)					Equipment Name:		QUENCH SEPERATOR				
Applicable Code:- ASME VIII DIV.1. 2023 ED					Equipment TAG No.		MV-PV021-V141004A/B, MV-PV022-V143004A/B				
Drq No. HEV000159-01 to 04-01					PO No:		36C/240840258		SQS Level / Criticality Rating : 1		
Mechanical Datasheet No. NP3701-MVVD-PV021-141004					Purchase Requisition: MT-N23P02-D11NPV-MVV002-TIN-P-01						
SR. NO.	ITEM DESCRIPTION	CHARACTERISTIC CHECK	EXTENT OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS/ STANDARDS	FORMAT OF RECORDS	AGENCY				REMARKS
							GMM	TPIA	tkUIPL	PMC/ RCML	
Legends: P=Perform, R=Review W= Witness, RW=Random Witness (Minimum 10%), H= Hold, M= Monitor, A=Approve, TPIA = Third Party Inspection Agency, RIL = Reliance Industries Ltd, DEC = Detail Engineering Contractor											
Notes :											
1	Material from China & East European countries not allowed. Material & Bought out items shall be procured from RIL Approved Sub vendors only.										
2	List of Items (Sourced within India) mentioned below shall be inspected by tkUIPL approved TPIA at Subvendor works as per approved TDC & EPCM approved surveyor.										
i	All Flanges above 24" (up to 300 # rating) for Carbon Steel, Low Alloy steel and Stainless Steel material.										
ii	All Flanges for 600 # rating and above.										
iii	All Body Flanges										
iv	Fasteners above M24										
v	All Fittings above 12" of Carbon Steel, Low Alloy steel and Stainless Steel material.										
vi	Formed components such as heads and tori cones										
vii	Critical internals such as demisters, distributors, coalescer pads etc										
viii	Final Painting (If off-loaded)										
ix	All Non ferrous and exotic material such as Duplex, Titanium, Monel, Inconel etc (Pipe, Fittings, Flanges etc.) irrespective of Size										
3	Scope & Extent of Inspection (H, W, RW, R, M) may vary based on Supplier Quality Surveillance level (SQS Level)										
4	TPIA shall perform Stagewise & Final Inspection as per approved ITP.										
5	Final Visual and Dimension inspection shall be done as per Code-1 approved Documents.										
6	Requirements of raw material, Inspection & Testing shall be as per RIL Specifications, Material Requisition, PO, Drawing & TDC.										
7	All temporary attachments shall be identified with P numbers and same shall be punched on part before attaching to pressure parts.										
8	Calibrated Instruments shall be used for inspection. Calibration Certificates of measuring instruments shall be made available to TPIA for review during inspection. TPIA shall verify calibration status of measuring instruments.										
9	All Tack Welds shall be carried out using qualified welders with qualified WPS.										
10	Protect Gasket sealing surfaces (i.e. serrations) during fabrication/Manufacturing from damages.										
11	Reinforcement pads beneath lifting lugs / trunnions shall be UT tested to verify absence of lamination in the steel.										
12	Studs shall exceed nuts by at least two threads and studs shall be threaded full length.										
13	Temporary attachments shall be removed (without hammering) and vessel wall examined by magnetic particle method (MT) or liquid penetrant method (PT).										
14	Longitudinal weld seams of adjacent shell courses shall be staggered to a minimum chord length of the greater of 150mm or five times the thickness of the thicker shell course.										
15	Circumferential seams shall be located to clear circumferentially welded internals such as tray support rings, vacuum stiffener rings etc. to avoid HAZ of weld and to facilitate NDT of circumferential joint.										
16	All Vessel seams shall be located to clear all external and internal permanent attachments. The distance from the edge of the attachment weld to the edge of main weld shall be at least 50mm or twice the thickness of the pressure part, whichever is greater. In any case reinforcing pads or any other pads shall not cover a welded joint.										

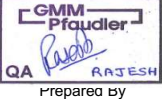
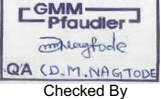
Inspection and Test Plan for QUENCH SEPERATOR



CUSTOMER :- Reliance Chemicals & Materials Limited					ITP NO:		HEV000159-01&04-10, Rev.0, Date: 20.02.2025			
PROJECT:- 1000 KTA VINYL CHLORIDE MONOMER (VCM)					GMM SO No:		HEV000159-01 to 04			
Detail Engineering Contractor(DEC):- thyssenkrupp Uhde India Private Limited (tkUIPL)					Equipment Name:		QUENCH SEPERATOR			
Applicable Code:- ASME VIII DIV.1. 2023 ED					Equipment TAG No.		MV-PV021-V141004A/B, MV-PV022-V143004A/B			
Drg No. HEV000159-01 to 04-01					PO No:		36C/240840258		SQS Level / Criticality Rating : 1	
Mechanical Datasheet No. NP3701-MVVD-PV021-141004					Purchase Requisition: MT-N23P02-D11NPV-MVV002-TIN-P-01					

SR. NO.	ITEM DESCRIPTION	CHARACTERISTIC CHECK	EXTENT OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS/ STANDARDS	FORMAT OF RECORDS	AGENCY				REMARKS
							GMM	TPIA	tkUIPL	PMC/ RCML	
17	NDE operators shall be certified ASNT-Level II as a minimum. Also, all NDE procedures shall be approved by ASNT Level-III and shall be qualified with demonstration.										
18	Root run where access is not available from back side shall be welded with GTAW process only. Additionally, full welding (all the passes) of the butt joints for nozzle sizes NPS 2 and less shall be done by GTAW.										
19	All testing shall be carried out in NABL lab only.										

Sr No.	List of Major Applicable Documents		
1	RPMS-ENGG-SPC-MS-002 & O24P01-PMCD-PBT-ME-001	Standard Specification for Unfired Pressure Vessels & its amendment	
2	RPMS-ENGG-SPC-MT-004	Standard specification for Welding and Procedure Quality Requirements for Suppliers	
3	RPMS-ENGG-SPC-PP-014	Standard Specification for Protective Coatings (Field and Shop Painting)	
4	RPMS-ENGG-SPC-PP-018	Positive material identification for Piping & Mechanical Component.	
5	RPMS-PROC-ITP-MS-001	Inspection and Test Plan for Pressure Vessels, Columns & Filters	

 Prepared By	 Checked By	
---	---	--